# Le Nghia Hiep

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## Summary

A Computer Engineering student with a sharp systems-oriented mindset and a passion for core technology research. I possess a strong technical foundation, which enables me to quickly adapt to and master new technologies to solve complex problems. Beyond technical expertise, I take pride in my strong communication, presentation, and teamwork skills.

## Education

**Hanoi University of Science and Technology (HUST)** — Hanoi, Vietnam
*Majoring in Computer Engineering - CPA 3.60* — *Sep. 2023 – Present*

**Phu Xuyen A High School** — Phu Xuyen, Hanoi, Vietnam
*Achieved 28.7 scores (Group A00) in the High School Graduation Examination* — *2020 – 2023*

## Experience

**Modern Computer Networks and Communications Technology Lab - NetLab** — Hanoi, Vietnam
*Research Assistant* — *April 2025 – Present*

- Research on AIoT in Healthcare
- Applying AI to solve real-world problems

## Technical Skills

**Language**: C/C++ (Embedded System), Python (Data processing & AI), Java, JS/HTML/CSS
**Embedded Systems & Hardware**: ESP32 development, Sensors integration
**Networking & Protocols**: IoT Protocols (MQTT, HTTP/Restful API), Network Basics (TCP/IP, UDP)
**Tools & Platforms**: Git/Github, Arduino IDE/PlatformIO
**English**: Intermediate (TOEIC Equivalent 650-700, enough to read and understand AI papers and documentation)

## Projects

**AI-Driven ECG Arrhythmia Classifier**
*ESP32, MQTT, Deep Learning, TensorFlow, Scikit-learn* — Full-stack AIoT project

- Hardware Design & Integration: Designed schematics and developed prototypes integrating ESP32 microcontrollers, AD8232, and MPU-6050 sensors; optimized power management systems using TP4056 charging circuits and LiPo batteries.
- Communication Protocols & Connectivity: Implemented real-time data streaming using the MQTT protocol over wireless networks to ensure stable, low-latency communication between devices and the cloud.
- Hybrid Deep Learning Architecture: Designed and implemented a hybrid 1D CNN-LSTM model to classify cardiac arrhythmias. Used 1D CNN for spatial feature extraction from raw ECG signals and LSTM to capture long-term temporal dependencies in heart rate variability
- Signal Processing & Optimization: Developed a preprocessing pipeline to handle raw signal noise (Baseline Wander, Powerline Interference) using digital filters. Conducted data segmentation and normalization on the MIT-BIH Arrhythmia Database to ensure high model generalization
- Device images: *Direct link to circuit image*
- Source code: *CNN-1D-and-LSTM-for-Arrythmia-Classification*

**To do list Website**
*React.js (Vite), Tailwind CSS, Node.js, Express.js, MongoDB, RESTful API, Vercel, Render* — Solo web project

- Developed and deployed a full-stack task management web application with a modern, responsive Glassmorphism UI using React and Tailwind CSS
- Implemented secure User Authentication and Authorization workflows using JSON Web Tokens (JWT) and bcrypt password hashing
- Engineered robust RESTful APIs with Node.js/Express to handle CRUD operations for user-specific tasks stored in MongoDB Atlas

- Built advanced frontend logic to sort, filter, and categorize tasks dynamically based on due dates (Today, Upcoming) and priority levels
- Integrated Google Gemini API to enable Natural Language Processing for automated task creation and intelligent sub-task decomposition from complex objectives
- Source code: *https://github.com/HiepLee1509/To-do-listtt*
- Domain (Due to Render's free tier limitations, the backend may require a cold start (∼50s) on the first request): *https://to-do-listtt.vercel.app/login*

**RAG-based University Query Chatbot**

*RAG, Python, LangChain, LLMs, FAISS, Streamlit* RAG Chatbot

- Data Pipeline Development: Engineered a robust pipeline to extract and preprocess data (including advanced chunking) from complex academic regulation PDFs
- Retrieval Optimization: Leveraged Recursive Character Text Splitting and Embedding Models to transform unstructured text into high-dimensional vectors, managed within a FAISS vector database
- Advanced RAG Implementation: Integrated Context Retrieval to provide automated source citations for every response, ensuring transparency and factual accuracy for legal and regulatory information
- Deployment: Developed and launched an interactive user interface using Streamlit to facilitate seamless student-bot interaction
- Source code: *AI-chatbot-for-HUST-students*

## Honors And Awards

- *Second Prize in the Student Creative Ideas Challenge - SoICT 2025*
- *Awarded Academic Achievement scholarship of SoICT*
- *Awarded Lotte Scholarship, SCG Sharing the Dream Scholarship 2023, 2024*